

A review of the immersion liquid cooling technology for high-performance data centers

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ABSTRACT

In the context of global digital transformation, the rapid expansion of data centers has intensified the challenge of high-power cooling. Traditional air cooling, with its limited efficiency, is increasingly inadequate to meet current demands. Immersion liquid cooling (ILC) has thus emerged as a critical research focus in data center thermal management, owing to its superior heat dissipation capabilities, excellent temperature uniformity, and energy-saving advantages. This paper provides a comprehensive analysis of ILC technology, examining coolant classification and selection criteria, the operating principles and performance of various liquid-cooling structures, and their practical applications. Two primary coolant types—oil-based and fluorocarbon-based—are evaluated, along with three main ILC system configurations: buoyancy-driven single-phase, pump-driven single-phase, and two-phase systems, each suited to different heat load scenarios. Performance improvement strategies, including passive and active enhancement techniques, are also explored. The research concludes that ILC has entered the stage of large-scale demonstration, significantly improving energy efficiency and exhibiting clear trends toward higher density and modularization. However, challenges such as high costs, operational complexity, and the lack of unified standards persist. Future efforts should prioritize technological innovation, cost reduction, and standardization to facilitate wider adoption and support the development of green, efficient data centers.

1. Introduction

With the acceleration of global digital transformation, data centers, as key infrastructure for massive data processing and storage, have witnessed an explosive growth in scale (Huang et al., 2023). The widespread deployment of high-density computing equipment has led to a sharp increase in the heat generated inside data centers, making the issue of thermal management increasingly prominent. It should be noted that excessively high temperatures will not only accelerate the aging process of electronic devices and shorten their service life (Huang et al., 2025), but also force the server systems to operate at a reduced frequency, which in turn results in computing power loss. Meanwhile, the additional cooling load incurred to cope with high-temperature environments will significantly increase the auxiliary power consumption of the refrigeration system (see Fig. 1), ultimately leading to a notable reduction in the energy efficiency of data centers. Against this backdrop, efficient and reliable thermal management technology has become a core requirement for ensuring the stable operation and enhancing the energy utilization efficiency of data centers. Therefore, research on data

center thermal management technology is of great practical significance for promoting the green and efficient development of data centers, ensuring the reliable and safe operation of information and communication systems, and serving as a key support for achieving the sustainable growth of the digital economy.

Faced with the growing heat load of data centers, traditional air-cooling methods can hardly meet the heat dissipation needs due to their low heat dissipation efficiency and a high propensity to cause local hotspots. To improve heat dissipation efficiency, liquid-cooling technology, such as cold plate type (Hou et al., 2026) and immersion liquid-cooling (ILC) type (Ni and Yu, 2024; Kalantarpour and Vafai, 2026) has emerged as an effective alternative. Different from the indirect cooling adopted by the cold plate type, ILC technology directly immerses electronic components in the coolant for heat dissipation. It has the advantages of compact structure, high cooling efficiency, and good temperature uniformity, thus providing a new solution to break through the bottleneck of traditional heat dissipation technology. So far, researchers and institutions at home and abroad have invested substantial resources in exploring ILC technology, covering coolant selection

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(Chakraborty et al., 2024; Chen et al., 2024), system integration design (Huang et al., 2025; Cakal and Sarioglu, 2025), performance impact mechanisms (Huang et al., 2025; Vagiakis et al., 2025; Rokonuzzaman et al., 2025), liquid-cooling structure optimization (Wang et al., 2025; Huang et al., 2023; Song et al., 2025) and application demonstrations (Li et al., 2025; Bukar et al., 2025; Chen et al., 2025). Universities and research institutions have played an important role in theoretical research: through simulations and experiments, they have conducted in-depth analyses of the heat transport mechanism and flow characteristics of ILC systems, providing a theoretical basis for the design of ILC systems. On the other hand, enterprises focus more on the practical application of technology and product development; they have actively launched various ILC products and carried out application demonstrations in practical engineering. Despite the existence of multiple literature reviews on the development of ILC technology, these studies usually only focus on one aspect or specific application of ILC. For example, Zhou et al. (2024) conducted research on the ILC technology of data centers, introduced and explained its current application status, and compared the cooling principles of single-phase ILC technology and two-phase ILC technology. Roe et al. (2022) introduced the current progress of ILC for lithium-ion batteries and discussed the factors affecting ILC performance. To the best of our knowledge, there is no literature that systematically describes the development status of ILC technology for data centers in accordance with the research framework of “coolant selection → liquid-cooling structure design → influencing factor analysis → performance enhancement method”. Moreover, the existing research on the cooling efficiency, safety, and reliability of ILC for data center applications is relatively scattered, and there is a lack of a comprehensive evaluation system. Therefore, further exploration is still needed in both the theoretical research and engineering application of ILC in data center thermal management.

To resolve the aforementioned problems, this paper aims to comprehensively review the research progress in data center thermal management, with a focus on the selection, design, application, and development of ILC technology. Firstly, it systematically introduces the key performance indicators, types, and respective characteristics of immersion coolants. Secondly, it outlines the structural design, influencing factors, and performance improvement approaches of existing ILC systems. Finally, it summarizes the practical application prospects and key challenges of ILC technology in data center thermal management. In conclusion, this paper will conduct an in-depth exploration of advancements in ILC technology for data centers and make a reasonable projection for its application prospects, which is instrumental in advancing the sustainability of the data center sector, improving energy utilization efficiency, and guaranteeing the reliable operation of information and communication systems.

2. Data center thermal management technology

To ensure their long-term operational safety and reliability, data centers require highly efficient and dependable thermal management technologies. In terms of operational stability, such technology can ensure that equipment operates in a consistently suitable temperature environment, avoid hardware failures and software errors caused by overheating, thereby maintaining the normal operation of business processes and preventing economic losses due to downtime. From the perspective of energy utilization efficiency, efficient thermal management can accurately control energy consumption for heat dissipation, reduce unnecessary energy waste, improve the ratio of energy converted into effective computing power, and lower operating costs. In terms of equipment service life, a stable and suitable operating environment can slow down the aging rate of electronic components, extend the service life of equipment, reduce the frequency of equipment replacement, and save hardware investment costs. In view of this, efficient thermal management conforms to the concept of green and sustainable development and promotes the transformation and upgrading of the data center industry toward low energy consumption and high reliability.

In the data center thermal management, air cooling, liquid cold plate, and ILC are three typical methods, and their principles are different, as illustrated in Fig. 2. The principle of traditional air cooling is to use devices such as fans to drive the flow of air, and dissipate heat via convective heat transfer between air (as a heat transfer medium) and the surface of electronic components. The liquid cold plate, on the other hand, involves installing a metal plate with coolant channels under or on the side of electronic components. The coolant circulates in the plate, and heat transfer is achieved through heat conduction between the coolant and the wall surfaces of electronic components, followed by convective heat transfer of the coolant itself. The primary mechanism of ILC is the complete immersion of electronic components in a dielectric fluid, which allows the coolant to absorb heat directly. The coolant used generally has good insulation properties. From the perspective of heat transfer, ILC demonstrates superior performance over traditional air and cold plate cooling methods. Air has inherently poor thermophysical properties, specifically low specific heat capacity and thermal conductivity, resulting in limited heat transfer capacity. Under the same conditions, when air removes the same amount of heat, it experiences a significant temperature rise, leading to thermal gradients and uneven heating of electronic components. Although the liquid cold plate uses coolant circulation, the coolant only contacts a part of the electronic components, giving rise to contact thermal resistance. Heat must first be conducted from the component surface to the liquid cold plate and then removed by the coolant, leading to a long heat transfer path and limited cooling efficiency. In ILC system, electronic components are in full contact with the coolant, eliminating contact thermal resistance and shortening the heat transfer path. The specific heat capacity and thermal conductivity of coolants are usually far better than those of air, enabling

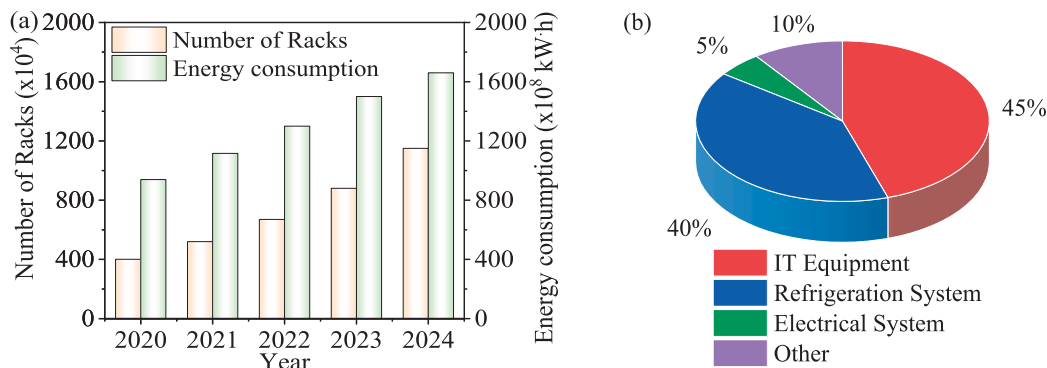


Fig. 1. Development trends and typical energy consumption structure of data centers in China: (a) Number of racks and electricity consumption; (b) Composition of electricity consumption.

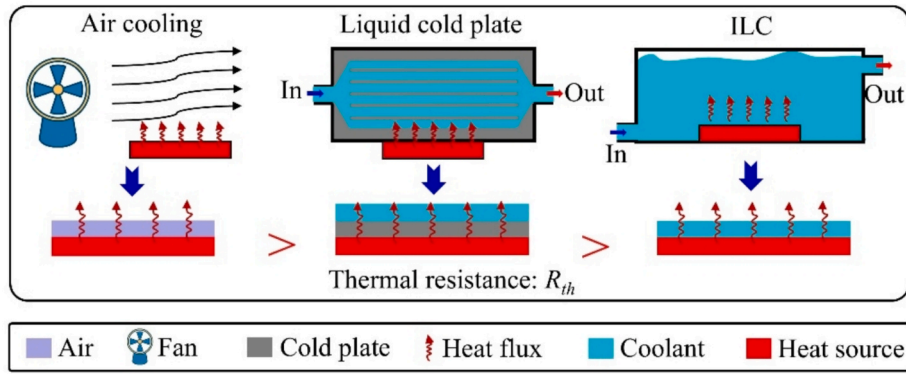


Fig. 2. Schematic diagram of different cooling methods.

them to absorb and remove heat more efficiently. Moreover, the coolant can be evenly distributed in the server, which helps enhance the thermal homogeneity of various components and greatly improves the overall thermal management effect. This is crucial for improving the performance of data center servers, extending their service life, and ensuring operational safety.

3. Advances and emerging trends in ILC technology

Due to the unique advantages of ILC technology, scholars have carried out a large number of theoretical and experimental studies in this field. Therefore, this paper focuses on coolant selection, liquid-cooling structure design, influencing factor analysis, and performance enhancement methods, aiming to comprehensively and systematically present the current development status of ILC technology.

3.1. Coolant selection profile

Given the direct contact between the coolant and electronic components, the choice of immersion coolant is of paramount importance to the performance and reliability of ILC systems. According to the differences in main components, the existing immersion coolants are primarily categorized into two groups: oil-based coolants and fluorocarbon coolants (Zhang et al., 2025), as shown in Fig. 3.

Among them, oil-based coolants mainly include organosilicon compounds and hydrocarbons, and can be further classified into mineral oils,

synthetic oils, and silicone oils. Fluorocarbon coolants can be divided into hydrofluorocarbons and perfluorocarbons according to their molecular structures. It should be noted that oil-based coolants (such as mineral oils and synthetic oils) have low costs, good insulation, and chemical stability, making them suitable for large-scale deployment. However, their thermal conductivity is relatively poor, and their viscosity is relatively high, which may lead to an increase in pumping energy consumption, and their relatively low flash points introduce a risk of combustion and explosion. Fluorocarbon coolants (such as perfluorocarbon liquids and hydrofluoroethers) have excellent thermal conductivity and insulation properties, and high heat dissipation efficiency, but they are expensive, and some products have poor environmental performance, which limits their large-scale application to a certain extent. Therefore, oil-based coolants are more suitable for scenarios with limited budgets and not extremely high heat dissipation requirements, while fluorocarbon coolants are more suitable for high-heat-density, high-performance computing high-end application scenarios. Table 1 presents some common immersion coolants in existing literature and their important properties.

In consideration of practical engineering applications, key properties of immersion coolants—thermal, electrical, chemical, and safety properties—must be holistically evaluated. In terms of thermal properties, different ILC systems have different requirements for the thermal properties of coolants. For buoyancy-driven single-phase ILC cases, a comparative study by Wang et al. (2023) on Novec 7100 and transformer oil revealed an inverse relationship between thermal conductivity and cooling efficacy. Paradoxically, despite a fivefold advantage in thermal conductivity, transformer oil exhibited a reduced immersion cooling capacity. The primary cause was identified as its elevated kinematic viscosity, which suppressed natural convective heat transfer by decreasing the Grashof number. In pump-driven single-phase ILC systems, Luo et al. (2022) prepared oil-based nanofluids by adding high-thermal-conductivity silicon carbide particles to mineral oil base fluids. As shown in Fig. 4, the thermal conductivity and dynamic viscosity of the composite coolant increase with a higher concentration of highly conductive nanoparticles, while its specific heat capacity gradually decreases. Given that thermal conductivity and specific heat capacity positively contribute to heat transfer enhancement, but an increased dynamic viscosity thickens the boundary layer and thus impairs heat transfer, a competition and coupling mechanism among these thermophysical properties is inevitable. Experimental results demonstrate that, despite the trade-offs, the surface heat transfer coefficient for the cooling of electronic components still achieved an overall enhancement of 11.7% with increasing nanoparticle concentration. This indicates that the heat transfer enhancement from the increased thermal conductivity outweighs the impairment caused by the higher dynamic viscosity and reduced specific heat capacity. For a two-phase ILC system, the influencing factors of thermal performance are more complex. In the analysis of data center two-phase ILC systems, Wu et al. (2025) found

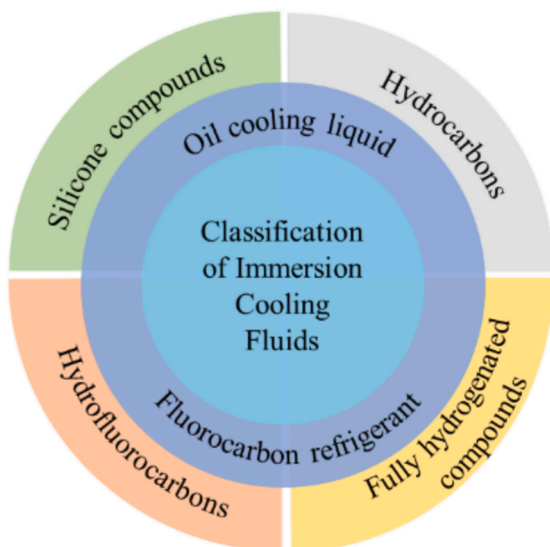
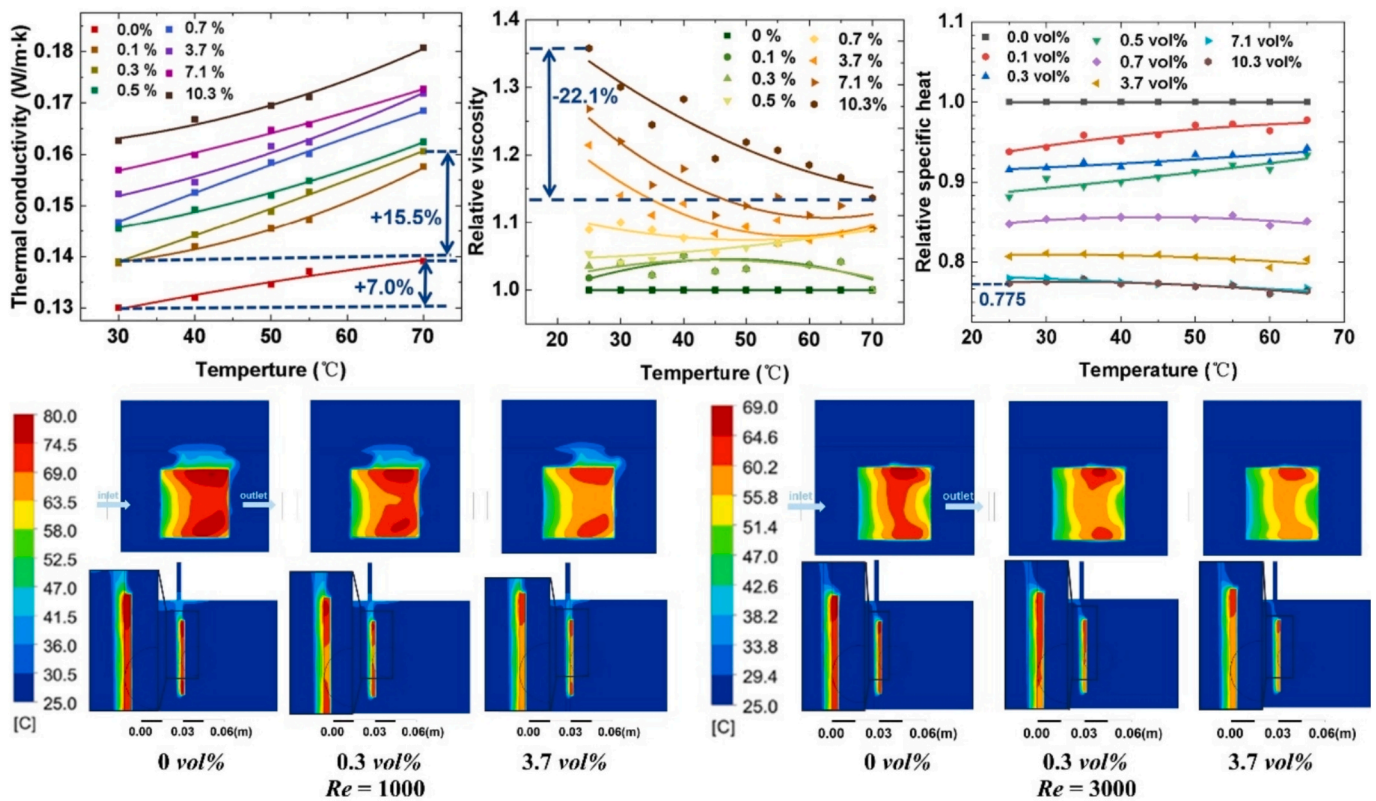


Fig. 3. Types of ILC coolants.

Table 1

Common ILC coolants and their important attributes (Zhang et al., 2025).

Name	Category	Boiling Point /°C	Specific Heat Capacity /J·g ⁻¹ ·K ⁻¹	Kinematic Viscosity /cSt	Thermal Conductivity /W·m ⁻¹ ·K ⁻¹	Liquid Density /kg·m ⁻³	Latent Heat of Vaporization /kJ·kg ⁻¹	GWP
Vertrel XF	HFC (25 °C)	55	1.13	0.42	0.065	1580	129.8	1600
Novec 7000	HFE	34	1.3	0.32	0.075	1400	142	530
Novec 7100		61	1.255	0.38	0.069	1420	112	320
FC-40	Perfluoroamine	165	1.1	2.2	0.065	1110	69	>5000
FC-70		215	1.1	12	0.07	1150	69	
HT55	Perfluoropolyethers(25 °C)	55	0.96	0.45	0.065	1530	92.1	
HFP-dimer	Perfluoroolefin (25 °C)	45.5–49.5	1.13	0.42	0.065	1580	98	<1
DFK			1.907	16.4	0.147	1520	88	
White mineral oil	Mineral oil	255–276	2.2	6.6	0.13	880		0
Dimethyl silicone oil	Silicone oil	170	1.758	5	0.15	915		0

**Fig. 4.** Trade-offs and interplay of coolant thermophysical properties (Luo et al., 2022).

that a lower boiling point of the coolant is beneficial to heat dissipation via latent heat of vaporization, that surface tension affects bubble behavior, and that these thermophysical properties also affect the stability of boiling heat transfer and system pressure. In their research on cooling high-heat-flux chips, Sun et al. (2022) employed Novec 7000 and Novec 649 under two-phase ILC conditions. Their findings demonstrated that Novec 7000, owing to its lower boiling point, could reduce the maximum chip temperature by 17.8 °C.

Furthermore, the direct contact between the coolant and electronics means that chemical properties, particularly material compatibility, have attracted significant research interest. Wen et al. (2016) conducted an experimental study on the material compatibility between the coolant and magnetic components in the two-phase ILC system for the first time. The results show that after the magnetic components are immersed in the coolant for a long time (≥ 1809 h), a maximum mass

change rate of merely 2.4% was observed, with a negligible influence of the coolant on the magnetic components' winding and leakage inductance. In addition, the electrical properties of a coolant represent a key parameter in determining its suitability. Research by Kurata et al. (2021) established an inverse relationship between a coolant's dielectric constant and the signal integrity of 25.8 Gbps silicon photonic micro-transceivers. Specifically, a lower dielectric constant resulted in a low bit error rate, and vice versa. At the same time, safety performance cannot be ignored. According to the findings of Karwa (2020), the application of specific immersion coolants could lead to ozone layer depletion and an aggravated greenhouse effect, and serious ecological problems may occur in case of leakage. An evaluation of these performance-influencing factors is central to the effective selection and application of coolants, and a series of evaluation indicators, as shown in Table 2, are needed to comprehensively evaluate the coolants.

Table 2
Evaluation indicators of ILC coolants.

Type	Indicators	Explanation	Standards
Thermal Performance	Figure of Merit (FOM)	Comprehensive evaluation of the heat transfer performance of coolants based on their thermophysical properties	(China Electronics Standardization Institute, 2024) (China Electronics Standardization Institute, 2023)
Electrical Performance	Dielectric Constant	It reflects the ability of the coolant to polarize and store electrical energy in an electric field (which affects signal transmission)	(Ministry of Industry and Information Technology of the People's Republic of China, 2021)
	Dielectric Strength	The critical electric field strength at which a coolant resists breakdown to form a conductive path (electrical safety margin)	
Chemical Performance	Material Compatibility	Characterizing the coolant's ability to inhibit irreversible degradation phenomena, such as chemical corrosion, swelling, dissolution, and embrittlement, during long-term coexistence with system contact materials	
Environmental Performance	Ozone Depletion Potential (ODP)	Quantifying the photochemical reaction efficiency of substances that destroy stratospheric ozone molecules after being released into the atmosphere	
	Global Warming Potential (GWP)	Quantifying the cumulative radiative forcing effect generated by substances absorbing infrared radiation and persisting in the atmosphere over a specific time scale	

3.2. Liquid-cooled structural design methodology

Within the domain of data center thermal management, the design of the ILC structure is the key to achieving efficient heat dissipation. Moreover, different types of ILC architectures have their own characteristics and are designed for diverse operational environments. At present, the main cooling structures of ILC data centers are divided into three categories, as shown in Fig. 5, namely buoyancy-driven single-phase ILC, pump-driven single-phase ILC, and two-phase ILC.

As shown in Fig. 5(a), the buoyancy-driven single-phase ILC system is a relatively simple cooling system, and its functionality is derived from the thermal expansion and contraction behavior of the coolant. Specifically, when the coolant absorbs the heat dissipated by electronic components, the temperature rises, leading to volume expansion and density reduction, thereby generating upward buoyancy. Under the action of buoyancy, the heated coolant flows upward, carries the heat to the

condenser at the top of the system for cooling, and then the coolant sinks again to form a natural convection cycle. Li et al. (2020) studied and compared the cooling capacity of the buoyancy-driven single-phase ILC system and the forced-convection air-cooling method. Their findings demonstrated a 23.4% reduction in the peak temperature of the lithium battery when using the buoyancy-driven single-phase ILC system, compared to a traditional forced air-cooling system. Matsuoka et al. (2017) researched the performance of the buoyancy-driven single-phase ILC system for servers through experiments and simulations. The power usage effectiveness (PUE) metric was introduced to characterize the energy efficiency of data centers. It is defined as the ratio of the total facility energy consumption to the energy consumed solely by the IT equipment, with a value closer to 1 indicating higher energy efficiency. The results show that the buoyancy-driven single-phase ILC system achieves a PUE of less than 1.04 in terms of energy consumption, while the minimum PUE of a traditional air-cooling solution is about 1.1, and the cold plate cooling has higher energy consumption due to the pump-driven coolant circulation. Compared to cold plate and air-cooling technologies, the buoyancy-driven single-phase ILC approach offers distinct advantages. However, a bottleneck arises because the heat transfer in natural convection is inherently tied to the coolant's flow velocity and thermal conductivity, which ultimately limits the overall cooling capacity of the system. Moreover, the slow flow rate of the coolant is prone to induce thermal stratification, resulting in uneven local temperature distribution and thus reducing the overall heat transfer efficiency.

To break through the heat transfer limit of natural convection, scholars have proposed a design scheme of a pump-driven single-phase ILC system. As depicted in Fig. 5(b), by introducing pump driving force to replace natural convection with forced convection mode, the coolant exhibits a significantly higher flow rate and enhanced heat transfer efficiency. In the pump-driven single-phase ILC solution, coolants absorb heat efficiently through forced convection when flowing through electronic components, and then release the heat through the condenser in the external loop to form an efficient cooling cycle. The key outcome of this design is its suitability for higher heat-density conditions, achieved through the effective elimination of thermal stratification and a major enhancement in overall cooling capability. Parida et al. (2012) researched and compared the cooling capacity of the pump-driven single-phase ILC and traditional forced air-cooling for server racks. Their results show that the cooling power of a typical air-cooled data center can reach 60% of the IT power, while the cooling power of the server using pump-driven single-phase ILC is only about 3.5% of the IT power. The pump-driven single-phase liquid cooling solution can reduce the cooling power consumption by more than 90%. In view of the shortcomings of the traditional indirect liquid cooling method using liquid cold plates, Choi et al. (2023) proposed a hybrid pump-driven single-phase ILC structure and studied its performance through experiments and simulations. The research results show that under the 3C charging condition, compared with the traditional bottom liquid cold plate cooling structure, the hybrid pump-driven single-phase ILC structure yields a 6.7 °C drop in the peak battery temperature and a 3.0 K decrease in the temperature span, effectively improving the batteries' thermal uniformity. Moreover, ILC methods can effectively control the battery temperature under various charging conditions, and are more able to meet the cooling needs of high-energy-density batteries compared with air cooling. Given this, the pump-driven single-phase ILC has significant advantages in thermal exchange efficiency and energy consumption compared with the buoyancy-driven single-phase ILC. However, due to the addition of pumps and complex circulation loops, the investment cost and later maintenance cost of the pump-driven single-phase ILC solution increase, and the risk of coolant leakage in the entire circulation loop is elevated, thus having higher requirements for material compatibility.

The core mechanism of the two-phase ILC system, which differentiates it from the other two structures, is a thermodynamic cycle of

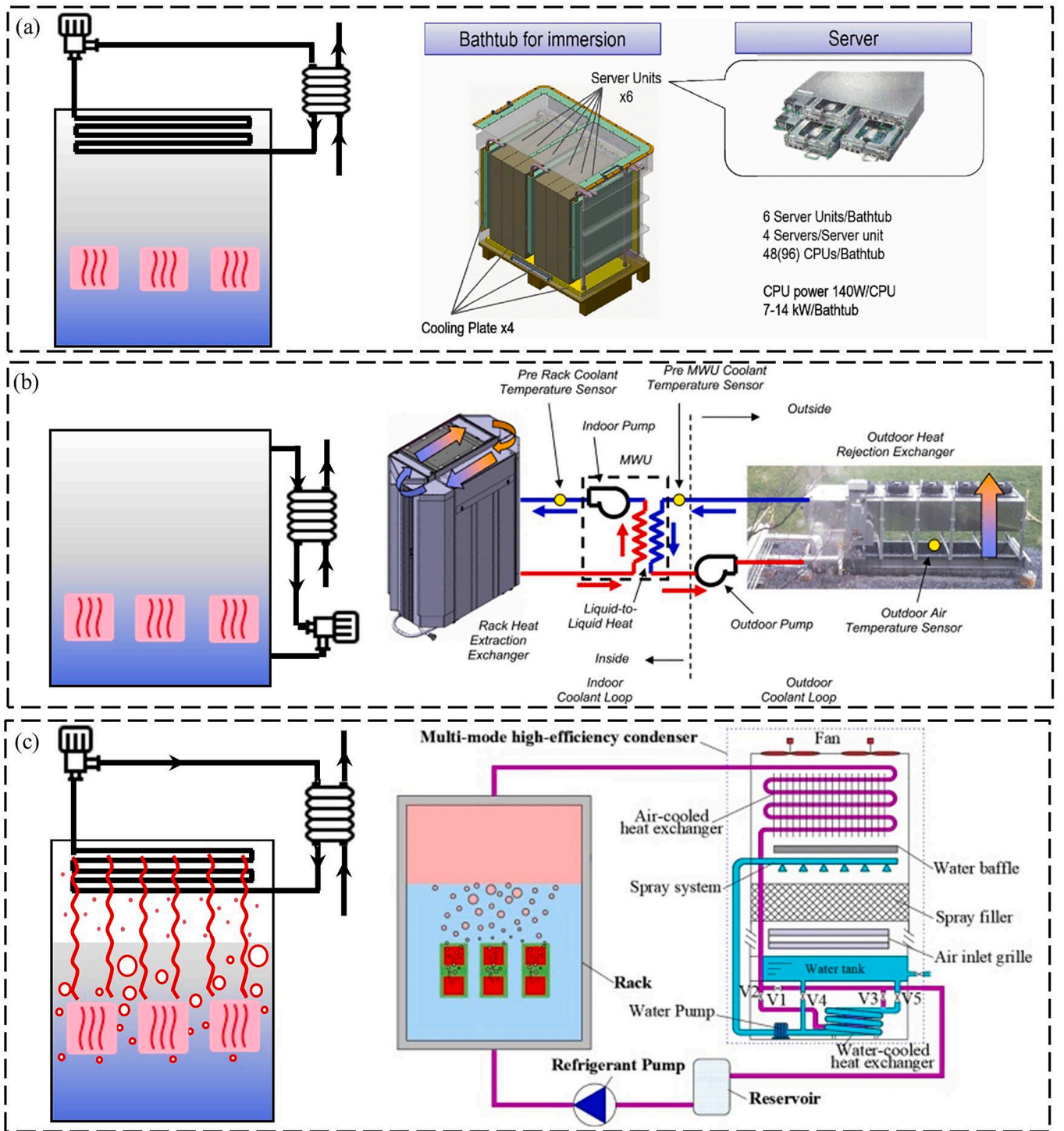


Fig. 5. Schematic diagrams of different ILC system frameworks. (a) Buoyancy-driven single-phase ILC (Matsuoka et al., 2017), (b) Pump-driven single-phase ILC (Parida et al., 2012), (c) Two-phase ILC (Zhang et al., 2023).

continuous evaporation and condensation. Specifically, as shown in Fig. 5(c), the coolant absorbs heat, boils, and transforms into vapor. This vapor separates from the vapor–liquid mixture and rises to create a gas-phase accumulation area above the liquid pool. Then, a condenser in the external circulation cools the vapor, making it condense back into a liquid, which re-enters the circuit in droplet form and continues to participate in the circulation. It can be seen that the two-phase ILC system effectively harnesses the coolant's latent heat, which helps to enhance the thermal exchange capacity and temperature uniformity of electronic components. Zhang et al. (2023) established a two-phase

immersion cooling system and calculated its operating performance. The results show that compared with the traditional liquid cold plate cooling and single-phase ILC methods, the two-phase ILC system offers a dual advantage: it ensures better server surface temperature uniformity by leveraging latent heat, and it exhibits significantly higher energy efficiency. Specifically, its coefficient of performance (COP) is 72–79% higher than that of single-phase systems at 6.6–15.9 kW, while operational data under loads of 3.43–9.17 kW show it can maintain a PUE = 1.15. In addition, the COP of the two-phase ILC system is much higher than that of the liquid cold plate cooling case, and the operating cost is

lower, which has significant advantages and energy-saving potential for data centers. Kanbur et al. (2021) built an experimental device for single-phase/two-phase ILC solutions, used commercial servers and ceramic heaters to simulate different power loads, and conducted a thermodynamic evaluation of the systems. The study concluded that the two-phase ILC system outperforms alternatives in both energy efficiency and cost-effectiveness. Quantitatively, it exhibits a 72.2–79.3% higher COP than the single-phase system at 6.6–15.9 kW. Economically, it holds a clear lifecycle cost advantage over the liquid cold plate system for servers with a power exceeding 5 kW. However, compared with single-phase ILC, the two-phase solution needs to maintain a high degree of sealing to ensure that the coolant does not leak, which not only increases the system's cost and complexity but also brings inconvenience to later maintenance.

In summary, flow and heat transfer mechanisms exhibit distinct differences across various cooling configurations in ILC systems. Huang et al. (2023) systematically elucidated the underlying physical

mechanisms of natural convection and forced convection in single-phase ILC processes. As illustrated in Fig. 6, forced convection significantly enhances the overall cooling performance of the system, while the presence of natural convection tends to exacerbate temperature stratification within the coolant, thereby compromising heat transfer uniformity. These findings demonstrate that the cooling configuration directly determines the refrigeration capacity and applicable scenarios of ILC systems, and must be carefully considered in practical applications. The thermal performance of the buoyancy-driven single-phase ILC system is easily affected by environmental factors, so it is more suitable for scenarios such as computing workstations and small server cabinets with low heat flux density. The pump-driven single-phase ILC solution has strong cooling capacity and can be applied to large-scale data centers with high-heat-load servers and high-density integration. In addition, the two-phase ILC system is mainly suitable for the new generation of intelligent data centers that are highly intelligent and have low maintenance requirements.

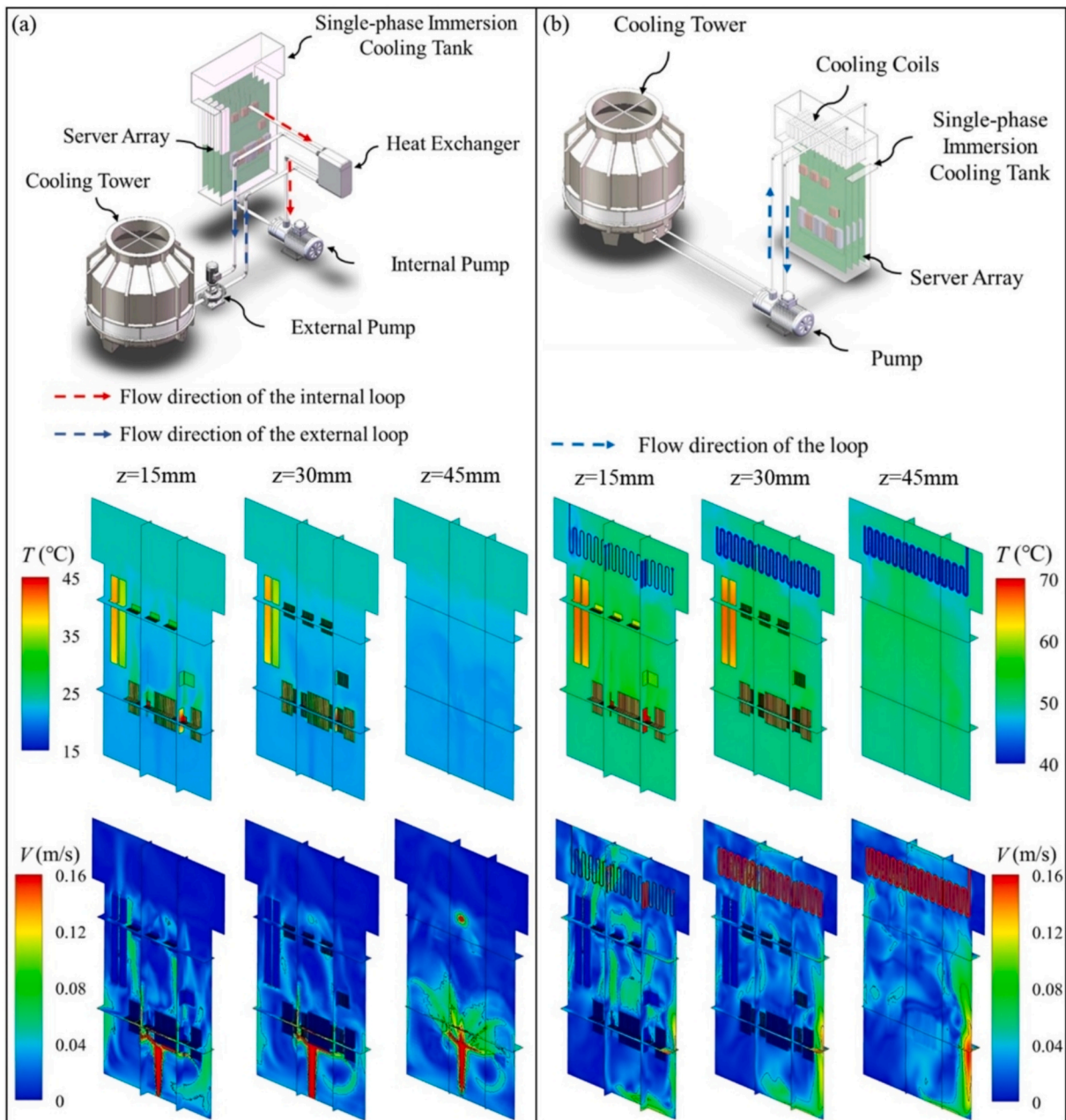


Fig. 6. Impact of different convective heat transfer mechanisms in single-phase ILC systems. (a) Pump-driven type, (b) Buoyancy-driven type (Huang et al., 2023).

3.3. Factors affecting system performance

For the single-phase ILC system, its performance is mainly affected by the thermophysical properties of the coolant (Liu et al., 2025; Kim et al., 2025); operating conditions (Chandrasekaran et al., 2025; Lyu et al., 2025); and liquid-cooling structure (Shen et al., 2024; Ozer, 2025). The cooling performance is influenced by the combined effect of the coolant's density, specific heat capacity, and thermal conductivity. For example, a large product of density and specific heat capacity means strong cooling capacity per unit volume, a high thermal conductivity is beneficial to heat transfer, but a high viscosity will increase the flow resistance loss and reduce the heat transfer capacity. Huang et al. (2024) experimentally compared the cooling performance of ILC systems using different electronic fluorinated liquids. They found that the coolant

exhibiting the lowest dynamic viscosity resulted in a 9.3% reduction in the maximum temperature and a 6.4% decrease in the system pressure drop, while also improving the PUE. As shown in Fig. 7(a), Hnayno et al.'s experiments (Hnayno et al., 2023) with coolants S5X, SmartCoolant, and ThermoSafe R established a clear inverse relationship between coolant viscosity and single-phase ILC performance. The study reported an approximately 6% reduction in cooling effectiveness corresponding to a viscosity increase from 4.6 mPa·s to 9.8 mPa·s. In terms of operating conditions, the thermal potential for heat dissipation is significantly affected by the coolant's inlet temperature, thereby determining the heat dissipation effect; a higher coolant flow rate generally enhances the heat dissipation capability but may also increase the pump power consumption. To characterize the thermal behavior of a single-phase ILC system, Cheng et al. (2020) employed a technique

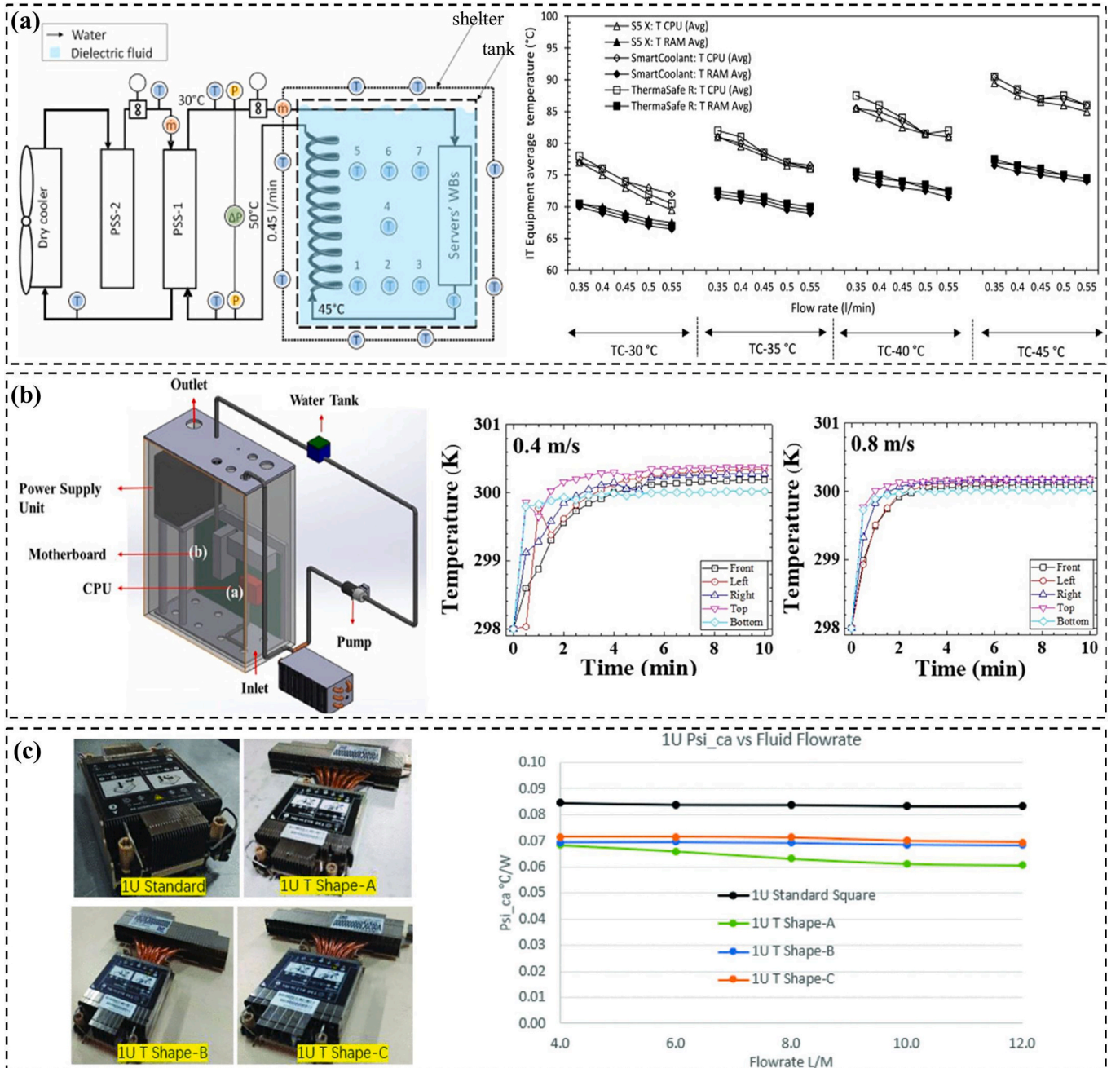


Fig. 7. Influence factors of single-phase ILC system: (a) Coolant types (Hnayno et al., 2023); (b) Operating conditions (Cheng et al., 2020); (c) liquid-cooling structure (Wang et al., 2024).

combining cross-sectional analysis and 3D isothermal surface reconstruction to observe heat distribution patterns at different flow rates. As depicted in Fig. 7(b), they found that an increase in the coolant flow rate can dissipate a greater amount of heat, resulting in a decrease in the CPU temperature, while a slow flow rate will lead to a higher temperature and uneven heat distribution. The objective of the experimental study by Li et al. (2024) was to determine the influence of inlet temperature, coolant flow rate, and server load ratio on single-phase ILC performance. Their results illustrated that reducing the inlet temperature of the coolant and the load ratio of the GPU, and increasing the flow rate are beneficial to reducing the average temperature of the GPU. In addition, in terms of the liquid-cooling structure, changing the size and shape of the heat dissipation fins can expand the heat transfer area and enhance convective heat transfer. Herring et al. (2024) evaluated the ILC performance of a new type of heat sink based on electrochemical additive manufacturing in their research. Compared to conventional parallel plate fins, the new heat sink's innovative geometry owns a more extensive specific surface area, leading to enhanced cooling performance in single-phase ILC applications within data centers. This aligns with the work of Wang et al. (2024), who investigated the performance of both 1U and 2U single-phase immersion-cooled servers. As illustrated in Fig. 7(c), their study found that for the 1U server, the T-Shape-A dense fin heat sink has the best performance, and the standard square heat sink has the worst performance. However, for the 2U server, the T-Shape-A dense fin heat sink has the best performance, but the standard square heat sink outperforms the T-Shape-B and T-Shape-C loose-fin designs.

Compared to single-phase ILC, two-phase ILC performance depends critically on the coolant's thermophysical properties and the component surface's cooling structure. In terms of the thermophysical properties of the coolant, the boiling point determines the temperature at which phase change occurs, and a lower boiling point can make the coolant

boil and absorb heat at a low temperature to enhance the heat dissipation capacity. The coolant's viscosity plays a critical role in governing flow dynamics and bubble evolution behavior. Furthermore, its heat removal capacity under a given mass is directly proportional to the latent heat of vaporization. As illustrated in Fig. 8(a), Li et al. (2021) compared three coolants (ethanol, FC-72, R113) for two-phase ILC performance. Ethanol demonstrated the lowest total thermal resistance, as low as $0.073^{\circ}\text{C}\cdot\text{W}^{-1}$ and $0.2^{\circ}\text{C}\cdot\text{W}^{-1}$ under forced air and natural cooling, respectively—a result directly linked to its maximum saturation temperature and latent heat of vaporization. For this boiling phase change heat transfer process, the morphology and structure of the heat dissipation surface are crucial. Generally speaking, in the two-phase ILC system, surface treatment methods such as microstructures (Li et al., 2024; Yan et al., 2025); nanostructures (Zhou et al., 2024; Jiang et al., 2024) and high-thermal-conductivity material coatings (Wang et al., 2024; Sinha et al., 2022) on the surface of electronic components can change the surface energy characteristics, increase the gasification nucleation sites, and thus improve the boiling heat transfer performance. As shown in Fig. 8(b), Mohammed et al.'s numerical analysis (Mohammed et al., 2025) demonstrated that coating server high-heat-flux components with copper reduced the maximum temperature to 65°C and improved inter-server temperature uniformity by 13% in a two-phase ILC system. Beyond the aforementioned classical methods, current research on surface modification is advancing toward the construction of multiscale composite structures, the introduction of intelligent responsive characteristics, and the utilization of novel low-dimensional materials. The core objective is to achieve more precise regulation of bubble nucleation, growth, and detachment behaviors during boiling processes, thereby further breaking through the heat transfer limit.

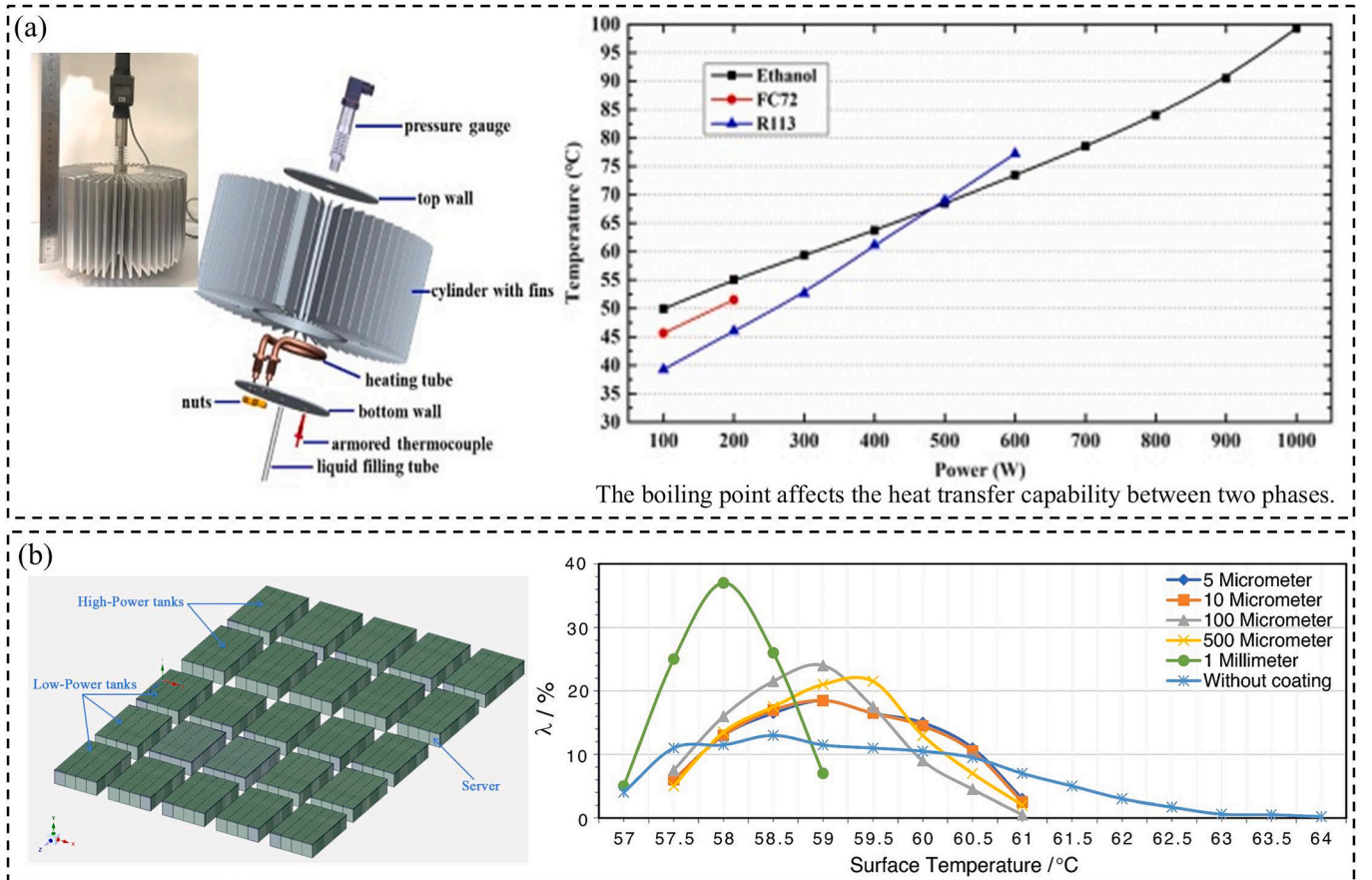


Fig. 8. Influence factors of two-phase ILC system: (a) Coolant types (Li et al., 2021); (b) Surface treatment methods (Mohammed et al., 2025).

3.4. System performance enhancement methods

Coolant modification, liquid cooling structure optimization, and interface regulation are the three mainstream technical pathways for enhancing the performance of immersion liquid cooling systems. It should be pointed out that the core objective of modifying the thermo-physical properties of coolants lies in tailoring its key parameters such as boiling point, latent heat of vaporization, wettability, and thermal conductivity. Moreover, the research, development, and application of this technology are intricately linked to the theories and technical foundations of materials science. Given that its underlying mechanisms have been systematically elaborated in previous sections, this chapter will shift its focus to analyzing other ILC enhancement methods, namely liquid cooling structure optimization and interface regulation.

Heat dissipation enhancement in single-phase ILC systems primarily centers on two strategies: increasing the heat transfer area and intensifying convective heat transfer between components and the coolant. Passive enhancement, in particular, is typically achieved through structural optimization of heat sinks and adjustments to the flow configuration. In terms of heat sink structure optimization, common methods include adjusting the thickness, number, height of fins, and the base plate's thickness, which significantly affects the cooling effect. Based on their research illustrated in Fig. 9(a), Sarangi et al. (Sarangi et al., 2022) proposed that a heat sink better suited for single-phase ILC requires thicker fins to boost efficiency and wider spacing to optimize convective cooling. In terms of adjusting the flow layout, the reasonable design of the inlet and outlet layout and internal flow channels is of great significance. Moreover, different inlet and outlet configurations will produce different heat transfer effects. A comparative analysis by Muneeshwaran et al. (2023) (Fig. 9b) on inlet/outlet configurations for single-phase ILC revealed the superior performance of the T-type layout, which reduced thermal resistance by 12.6% and the maximum temperature by 2.8 °C compared to the Z-type. In addition, the optimization goals of the flow channel design of the single-phase ILC system include

uniform flow distribution, enhanced fluid disturbance, and on-demand flow redistribution (Choi et al., 2023; Liu et al., 2021); which also have a significant impact on enhancing the heat dissipation effect. As illustrated in Fig. 9(c), Ge et al. (2024) compared the benefits of optimizing the coolant flow distribution by using a diversion baffle in a numerical study. The results show that adding a baffle in the inlet part of the system effectively improves the ILC performance, and it is crucial to reduce the bypass channel and optimize the coolant flow distribution to maximize the heat dissipation capacity. Liu et al. (2024) designed a single-phase ILC structure using a combination of a baffle and a porous panel, which ensures the coolant's uniform distribution in servers and reduces the coolant bypass, thereby greatly reducing the peak temperature and temperature delta of electronic components. To further improve the performance of single-phase ILC, scholars have also proposed active heat transfer enhancement methods relying on jet assistance, as shown in Fig. 9(d). Yuan et al. (2024) researched the heat dissipation performance of a jet-assisted ILC system through experiments. Experimental data revealed a 2.6-fold enhancement in the steady-state surface heat transfer coefficient for the jet-assisted ILC system over the conventional single-phase system, reaching a peak value of $1745.0 \text{ W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$. In addition, Liu et al. (2025) also conducted a similar comparative study, and they found that the temperature of electronic components and the coolant's temperature uniformity index in jet-assisted single-phase ILC systems decreased by 6.1% and 73.1% respectively. This phenomenon results from the enhanced local turbulent disturbances in the high-heat-flux region caused by jet impingement, which significantly improves the convective heat transfer efficiency. While passive methods, including heat sink optimization and flow channel improvement, currently form the basis for enhancing convective heat transfer in single-phase ILC systems, the exploration of active performance enhancement techniques like jet impingement remains limited, underscoring a gap in current research.

For the two-phase ILC system, the performance enhancement strategies of the single-phase ILC system are also effective. Driven by the

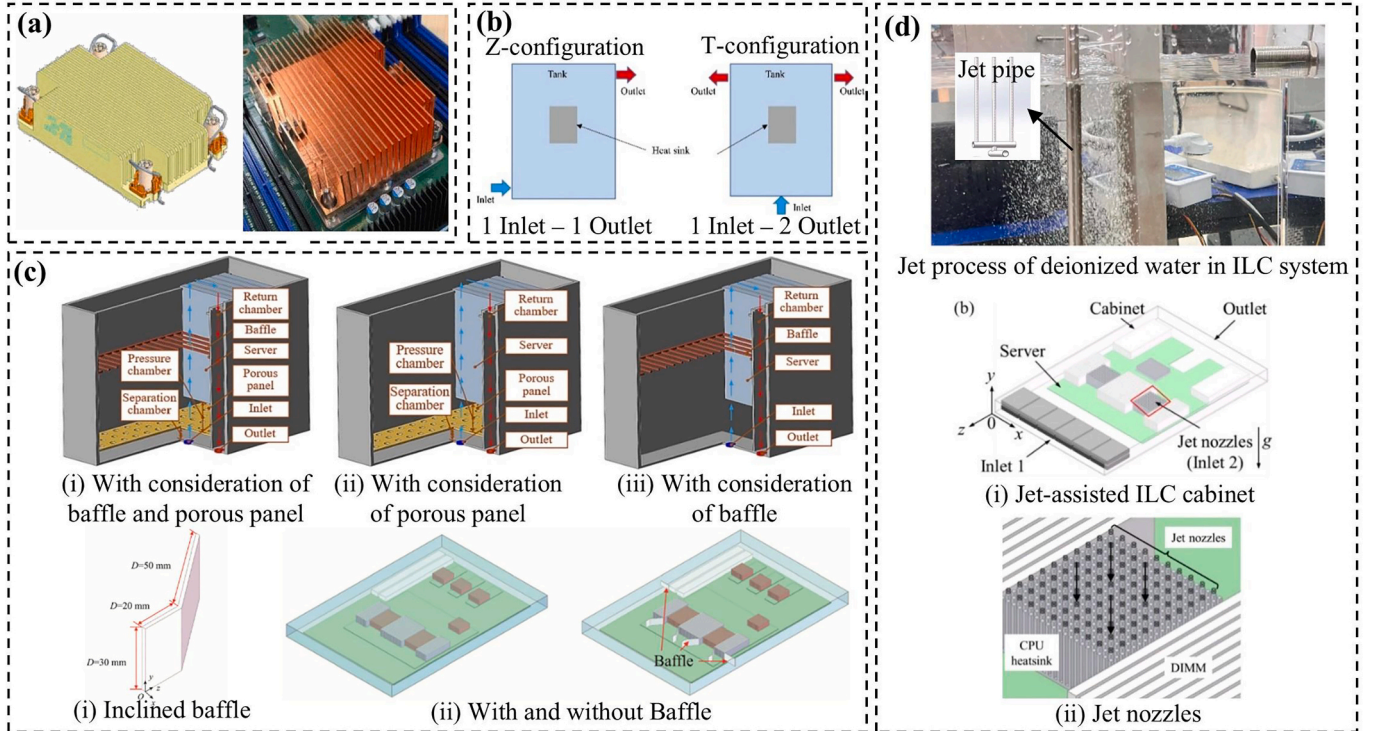


Fig. 9. Enhancement technologies for single-phase ILC: (a) Optimization of fin size (Sarangi et al., 2022); (b) Optimization of inlet and outlet layout (Muneeshwaran et al., 2023); (c) Optimization of flow channel design (Ge et al., 2024; Liu et al., 2024); (d) Enhancement assisted by jet impingement (Yuan et al., 2024; Liu et al., 2025).

need for more efficient heat removal from data center servers, Zhou et al. (2022) studied a two-phase ILC strategy using a vapor chamber heat sink, and innovatively adopted a gradient capillary wick structure in the vapor chamber. The results show that when the heat source temperature is lower than 85 °C, the vapor chamber can effectively transfer a heat load of 500 W. Sun (2022) conducted a numerical simulation calculation on the phase-change process of the two-phase ILC cabinet. The study identified a primary dependency of cabinet temperatures (maximum and average) on chip spacing, which showed a gradual decrease. A more complex, non-monotonic dependence was observed with inlet temperature, characterized by an initial decrease and subsequent increase in cabinet temperature. Under a certain set working condition, there will be an optimal inlet temperature value. Considering that the core principle of two-phase ILC is that the coolant takes away heat through boiling phase change on the thermal exchange surface, the research on its performance enhancement focuses on

enhancing the boiling heat transfer performance. At present, there are three passive enhancement methods for the performance of the two-phase ILC system, namely surface treatment (Shamsaiee et al., 2022; Gajghate et al., 2021); addition of porous foam (Pranoto et al., 2012; Sharifzadeh et al., 2022) and surface modification (Li et al., 2020; Mehrizadeh et al., 2020). Among them, the most common method is to modify and customize the design of the heat transfer surface structure, as shown in Fig. 10(a). Ali (2020) evaluated the thermal performance of flat and micro-porous copper heat sinks in a two-phase ILC system through a numerical comparison. This study shows that the total thermal resistance of the micro-porous copper heat sink is smaller than that of the traditional flat copper heat sink, which greatly reduces the peak temperature, significantly lower than the temperature limit set by the high-power chip industry. Chuang et al. (2021) used metal 3D printing and injection molding technology to custom-design an integrated heat sink, which can enhance the boiling characteristics of two-phase ILC and

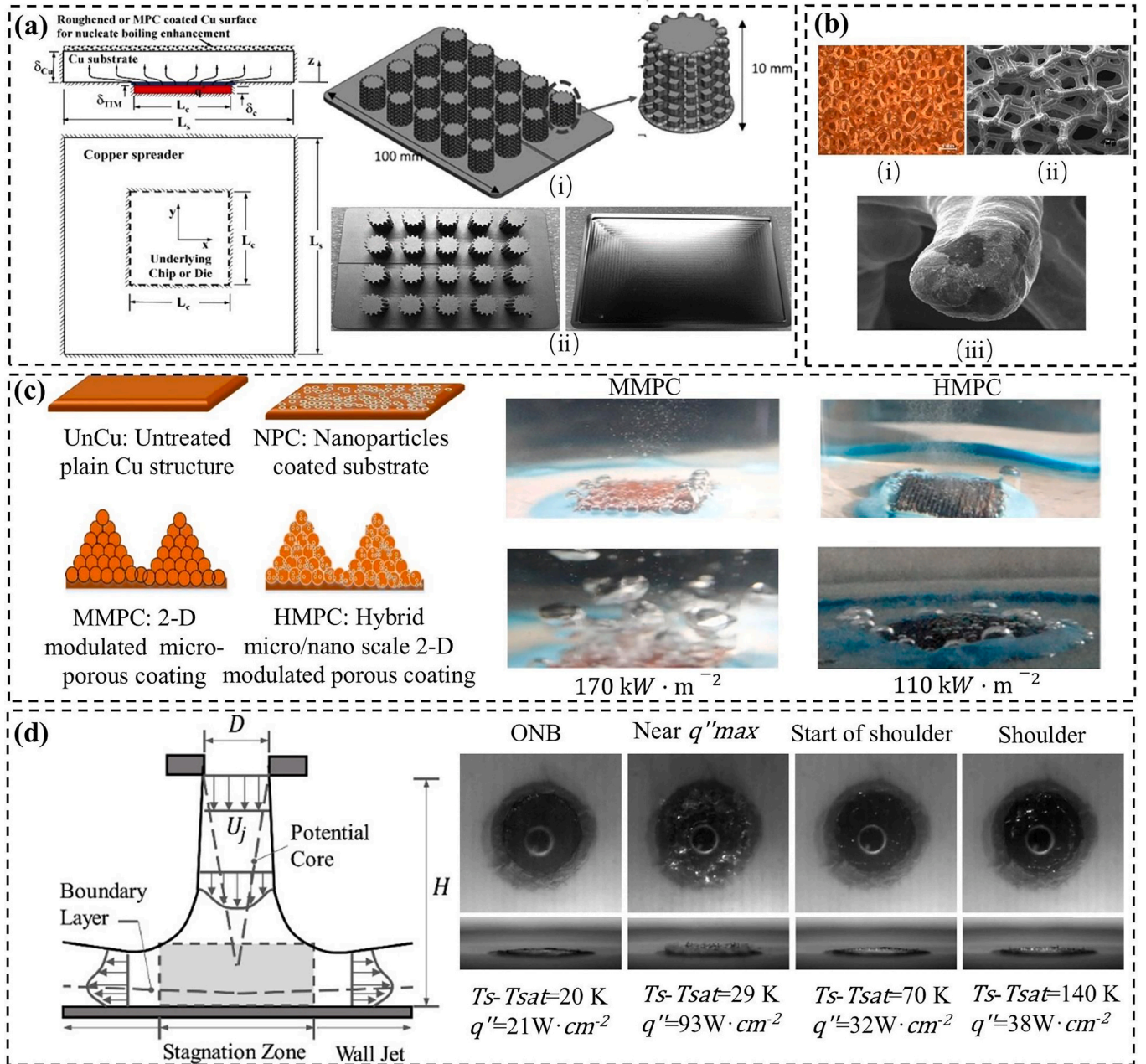


Fig. 10. Enhancement technologies for two-phase ILC: (a) Surface treatment (Ali, 2020; Chuang et al., 2021); (b) Addition of porous foam (Manetti et al., 2020); (c) Surface modification (Khan et al., 2019); (d) Assistance of jet impingement (Stamps and Weibel, 2024).

improve the cooling performance by more than 50%. Adding porous foam to the heat transfer surface is also an effective method. A key feature of the foam's 3D porous architecture is its ability to markedly enhance the density of nucleation sites. As shown in Fig. 10(b), Manetti et al. (2020) conducted pool boiling experiments with three different thicknesses of copper foam (3 mm, 2 mm, and 1 mm) in a two-phase ILC system with HFE-7100 as the coolant. Optimal performance was achieved under low heat flux with 2 mm or 3 mm copper foam. In contrast, under high heat flux, the 1 mm foam exhibited the highest boiling heat transfer capacity. In addition, the surface modification structure of the heat transfer surface can also play a critical role in determining its boiling heat transfer performance. As plotted in Fig. 10(c), Khan et al. (2019) examined the behavior of various surfaces during pool boiling within a two-phase ILC system. A comparative study with ordinary surfaces found that the convective thermal transport coefficient and critical heat flux of the nano-coated copper surface are improved, and the performance of the porous coated surface is lower than that of the micro-porous surface. Compared with passive heat transfer enhancement, the current research on active enhancement of two-phase ILC is still in its infancy. Jet impingement is a technology to enhance flow boiling. Compared with pool boiling, it can significantly increase the critical heat flux with little additional pump power consumption. As shown in Fig. 10(d), Stamps and Weibel (2024) proved through experiments the occurrence of the shoulder effect in the immersion jet impingement in the two-phase ILC system, and discussed the dependence of the shoulder heat flux on the inlet subcooling and jet velocity. Their research shows that the jet impingement two-phase ILC has efficient heat transfer capacity for high-heat-flux hotspots, making it one of the most solutions for electronic thermal management. In conclusion, the advancement of two-phase ILC centers on optimizing surface-coolant interactions through surface engineering (e.g., treatments, porous media, modifications) to enhance boiling, coupled with active forcing mechanisms like jet impingement to elevate the heat removal ceiling.

4. Application of ILC technology in data centers

4.1. Engineering demonstrations

At present, the global data center ILC technology has entered the stage of large-scale application demonstrations, and a variety of technical routes are developing in parallel, which has significantly improved the energy efficiency of data centers. Table 3 summarizes the current domestic and foreign ILC products and data center application demonstrations. In China, Alibaba Zhejiang Cloud Computing Renhe Data Center adopts pump-driven single-phase ILC technology, deploys 5100 racks (100,000 servers), and achieves a PUE as low as 1.09. Zhejiang Juhua Data Center verifies the feasibility of PUE = 1.07 in small-and medium-scale scenarios through 8×25 kW single-phase liquid-cooled racks (with a total load of 200 kW). Zhongke Shuguang "Silicon Cube" Changsha 5A Intelligent Computing Center adopts a more efficient two-phase ILC technology, with a single rack power density of 220 kW and a PUE further reduced to 1.04, highlighting the technical advantages in high-density computing scenarios. Internationally, Microsoft Azure Data Center (USA) and Intel G-Flow Project achieve PUE < 1.1 and 1.05 through two-phase/single-phase oil ILC respectively, while the modular liquid cooling solutions (22–60 kW per rack) of Green Revolution Cooling (USA) and Asperitas (Netherlands) promote edge computing applications with flexible deployment capabilities, and the PUE is generally controlled between 1.05 and 1.1. These cases show that ILC technology has moved from the experimental stage to commercialization. Single-phase solutions are easier to popularize due to their low cost, while two-phase technology shows potential in extreme scenarios such as supercomputing/AI. Overall, it presents three major trends: "high energy efficiency (PUE generally below 1.1), high density (single rack exceeding 200 kW), and modularization (rapid deployment)".

Table 3

Domestic and foreign ILC products and application demonstrations in data centers.

Project Name	Location	Type of ILC	Scale of Data Center	Performance Indicators
Alibaba Zhejiang Cloud Computing Renhe Data Center (Stamps and Weibel, 2024)	Hangzhou, China	Pump-driven single-phase ILC	5100 standard racks, equipped with 100,000 servers	PUE = 1.09
Zhejiang Juhua ILC and Data Center (Overview of Major National Data Centers, 2025)	Quzhou, China	Pump-driven single-phase ILC	8×25 kW liquid-cooled racks, with a maximum total load of 200 kW, including 138 servers (192U)	PUE = 1.07
Zhongke Shuguang "Silicon Cube" Changsha 5A Intelligent Computing Center (The "Calm Guardian", 2024)	Changsha, China	Two-phase ILC	220 kW per rack	PUE = 1.04
Microsoft Azure Data Center (The "Calm Guardian", 2026)	Washington State, USA	Two-phase ILC		PUE < 1.1
Intel G-Flow ILC (Wangsu Technology's Green Cloud Map, 2024)	USA	Single-phase oil-ILC	Kilowatt-level	PUE = 1.05
Green Revolution Cooling ICETank Modular Data Center (ICETank Modular liquid immersion cooling system, 2026)	USA	Single-phase ILC	60 kW per rack	PUE < 1.05
Asperitas AIC24 Modular Data Center (Netherlands' Asperitas Launches Liquid-Cooled Modular Data Center, 20178)	Amsterdam, Netherlands	Single-phase oil-ILC	22 kW per rack, 48 servers	PUE < 1.1

However, large-scale promotion is still restricted by challenges such as the cost of immersion coolants and the reconstruction of the operation and maintenance system.

4.2. Application prospects

With notable advantages such as efficient heat dissipation, energy conservation, environmental protection, and quiet operation, ILC technology exhibits extensive application potential in data centers, encompassing General-Purpose, Intelligent Computing, and Edge Computing Data Centers.

As shown in Fig. 11, General-Purpose Data Centers (GPDCs) are characterized by large scale, moderate equipment density, and stable services, with core demands of minimizing operating costs and meeting stringent PUE assessment requirements. ILC technology can eliminate

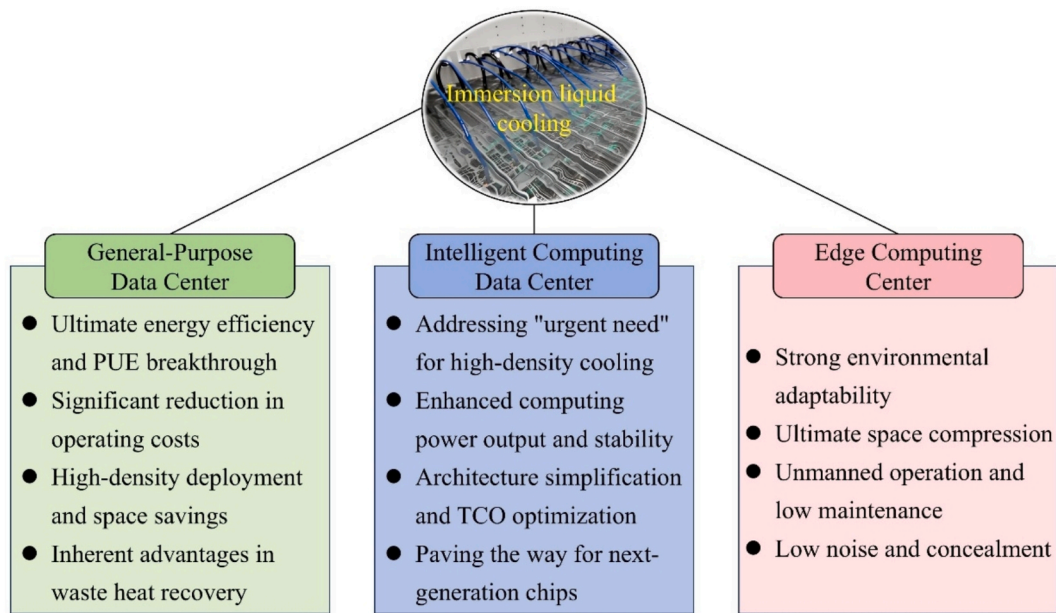


Fig. 11. Application prospects and advantages of ILC technology.

server fan energy consumption and most of the air conditioning refrigeration energy consumption, while the system's pumping energy consumption is extremely low—its PUE can easily drop below 1.1, far outperforming traditional air-cooled data centers. For large-scale GPDs, this translates into annual electricity cost savings of millions or even tens of millions of yuan, directly aligning with the green and low-carbon requirements of national strategies such as the “Eastern Data and Western Computing” project in China. In addition, it can significantly reduce water costs (eliminating the substantial water consumption of cooling towers or evaporative cooling systems) and maintenance costs (reducing the operational and maintenance workload of air conditioning systems). Although the power density per cabinet of GPDs is generally lower than that of Intelligent Computing Data Centers (ICDCs), ILC enables higher-density server deployment in the same space, or boosts the IT equipment installation capacity by reducing the space occupied by cooling infrastructure—thereby enhancing the overall computing capacity of the data center. More notably, the heat generated by the ILC system can be converted into high-quality hot water at temperatures above 50–60 °C, which is highly suitable for waste heat recovery scenarios such as district heating and agricultural breeding. This creates additional revenue for data centers and facilitates their true transformation into “Energy Hubs”.

ICDCs feature distinct scenario characteristics: ultra-high computing density (up to several tens of kW or even over 100 kW per cabinet), with GPUs as the core heat-generating components, explosive growth in computing power demand, and stringent requirements for computing stability—tasks such as large model training cannot be interrupted once initiated. Currently, the power consumption of a single GPU has exceeded 700 W and will continue to rise in the future, while traditional air-cooling technology has approached the heat dissipation limit (approximately 40 kW per cabinet), failing to meet the demands. ILC technology, however, can easily handle power densities of 100 kW per cabinet or higher, providing stable heat dissipation for high-performance GPUs and AI chips to ensure their continuous full-load operation without frequency reduction. More importantly, GPUs can maintain maximum boost frequency in the low-temperature environment created by liquid cooling, effectively avoiding performance throttling caused by high temperatures. For time-consuming training tasks lasting weeks or even months, this feature eliminates task interruptions due to heat dissipation issues, significantly improving

computing efficiency and operational reliability. In addition, ICDCs generally face the challenge of tight construction schedules. ILC adopts a modular and integrated design (e.g., cabinet-level or tank-level deployment), enabling rapid deployment and eliminating the need for constructing complex cooling infrastructure in traditional data centers. Although its initial investment may be higher, from the perspective of the whole lifecycle (TCO), the returns from substantial electricity cost savings and improved computing power output are extremely significant. Facing future CPUs/GPUs with power consumption exceeding 1000 W, ILC is by far the most effective and feasible heat dissipation solution, laying a solid foundation for the technological advancement of intelligent computing infrastructure.

Additionally, Edge Computing Centers (ECCs) are mostly deployed in harsh environments such as factory workshops, base station peripheries, and shopping mall basements, and they generally face core pain points including space constraints, a lack of professional IT operation and maintenance (O&M) personnel, and high sensitivity to footprint and noise levels. ILC systems precisely address these needs: (1) Their sealed structure is inherently immune to contaminants like dust, humidity, salt spray, and particulate matter in the external environment, enabling deployment in harsh scenarios that traditional data centers cannot tolerate and truly achieving “deployability anywhere”. (2) For space scarcity, the system can be designed into a highly integrated “cabinet-as-a-data-center” form by eliminating the bulky air conditioning system, it significantly reduces floor space. Moreover, it supports flexible wall-mounted or containerized deployment to fit various narrow spaces. (3) In addition, the system operates stably and reliably, featuring “maintenance-free” or “low-maintenance” properties. This is particularly critical for edge scenarios lacking professional O&M capabilities, as it substantially lowers O&M thresholds and costs. (4) Combined with its extremely low operational noise, it can seamlessly integrate into noise-sensitive areas such as offices, shopping malls, and hospitals without causing any disruptions.

4.3. Key challenges and opportunities

ILC technology currently faces a series of severe challenges on its path toward large-scale application, as illustrated in Fig. 12. In terms of investment costs, the high expense of specialized coolant and substantial infrastructure renovation fees form a significant initial capital

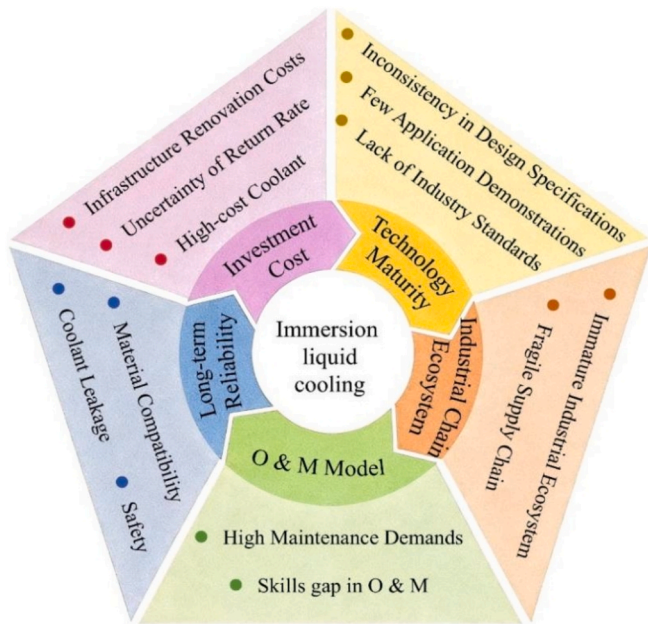


Fig. 12. Challenges facing the commercialization of ILC technology.

expenditure, creating considerable uncertainty regarding return on investment. The lack of technological maturity is reflected in the absence of industry-wide standards, inconsistent design specifications among manufacturers, leading to poor interoperability, and a relative scarcity of large-scale demonstration cases, indicating that practical experience still needs accumulation. Regarding long-term reliability, the prolonged compatibility between the coolant and various materials inside servers remains a key concern, while potential coolant leakage poses an ongoing threat to system safety. The operational model is undergoing a fundamental transformation: traditional maintenance personnel face a skills gap, and the complex, highly demanding “blind operation” maintenance procedures for equipment increase both the difficulty and cost of operations. Finally, the entire industry chain ecosystem remains fragile, with the supply chain concentrated among few suppliers and weak risk resilience. Coupled with an underdeveloped industrial ecosystem that has yet to form a healthy, collaborative development pattern, these factors collectively hinder the rapid adoption of this technology.

5. Conclusions

As a key infrastructure of the digital economy, data centers attach great importance to their thermal management technologies. ILC technology has become a current research and application hotspot owing to its superiority in excellent heat dissipation performance and good temperature uniformity. This paper comprehensively examines the development of ILC technology for data center thermal management, covering aspects such as coolant selection, liquid-cooling structure design, influencing factor analysis, performance improvement approaches, and engineering applications, and also discusses its application prospects and challenges. The main findings of this study are summarized as follows:

- (1) The selection of ILC coolants needs to comprehensively consider multiple factors such as thermophysical properties, electrical properties, chemical compatibility, and safety and environmental protection. At present, the mainstream coolants mainly include two categories: traditional oil-based coolants and fluorocarbon coolants. The former boasts distinct cost advantages but carries safety risks stemming from insufficient thermal conductivity and flammability; the latter has excellent heat dissipation efficiency

but is restricted by its high cost and the subpar environmental performance of some products. Driven by both high-performance requirements and environmental protection regulations, immersion coolants are developing in two directions: one is the development of environmentally friendly fluorocarbon coolants with low GWP, and the other is the performance optimization and upgrading of traditional oil-based coolants.

- (2) A tripartite classification of data center ILC structures exists, encompassing buoyancy-driven single-phase, pump-driven single-phase, and two-phase variants. The buoyancy-driven single-phase ILC structure is simple but fails to provide adequate cooling capacity, making it well-suited for scenarios with low heat dissipation requirements; the pump-driven single-phase ILC offers an ideal solution for high heat density situations due to its outstanding heat transfer efficiency, but incurs higher costs and maintenance complexity; the two-phase ILC uses the latent heat of the coolant, demonstrating good heat transfer and temperature uniformity performance, and is suitable for high-end data centers, but has high sealing requirements, high cost, and inconvenient maintenance.
- (3) The primary strategy for improving single-phase ILC performance relies on passive techniques, including heat sink structure optimization and flow layout optimization. In contrast, active enhancement approaches like jet-assisted ILC remain relatively unexplored. Furthermore, while two-phase ILC systems can utilize the optimization strategies of single-phase systems, their performance enhancement depends critically on boosting boiling heat transfer via surface treatment, porous foam integration, and modified surface structures. Active methods such as jet impingement also demonstrate significant potential in this area.
- (4) ILC technology has entered the stage of large-scale application demonstrations in data centers, which has significantly improved energy efficiency and presents the development trends of high energy efficiency, high density, and modularization. However, its large-scale application faces challenges such as high initial investment cost, complex operation and maintenance, poor compatibility with existing infrastructure, the need for improved coolant performance, and a lack of unified standards. In the future, technological breakthroughs are needed in technology research and development, cost control, and standardization to promote the wide application of this technology in the field of data centers and realize the green and efficient development of data centers.

6. Future perspectives

In summary, immersion cooling has emerged as a pivotal solution for supporting the development of high-performance data centers. Driven by the dual forces of the intelligent digital economy and the global green agenda, its future development will focus on three core evolutionary paths: intelligent management and control, system-level integration, and full-lifecycle environmental sustainability.

First, with the rapid advancement of artificial intelligence, thermal management research in immersion-cooled data centers is set to transcend traditional static paradigms, shifting toward the development of intelligent, adaptive systems powered by AI and the Internet of Things. To this end, it will be necessary to deploy embedded sensor networks that collect real-time operational parameters—such as temperature and pressure—along with coolant properties, thereby building a comprehensive performance database for multiple operating conditions. On this basis, machine learning algorithms can be employed to dynamically optimize key operational variables—including flow rate, pump power, and cooling water temperature—enabling predictive thermal management and adaptive control capabilities. This transition from passive response to proactive optimization will ultimately achieve an optimal synergy between energy efficiency and system performance. Core

challenges in this direction include the development of high-fidelity, robust digital twin models and the formulation of efficient machine learning algorithms capable of multi-objective cooperative optimization.

Second, the full potential of immersion cooling technology hinges on system-level and holistic innovation, moving beyond localized optimization of individual components. Future research should prioritize the co-design and deep integration of IT hardware and cooling infrastructure systems. Furthermore, there is an urgent need to explore new system architectures that closely couple immersion cooling with other data center subsystems—such as power supply and spatial environmental systems. Examples include developing low-energy, high-efficiency heat exchange technologies that repurpose waste heat for practical applications such as space heating or seawater desalination. This requires a fundamental understanding of the coupling mechanisms among fluid dynamics, thermodynamics, and other multi-physics phenomena, with the aim of constructing highly integrated, high-density, and energy-efficient data center modules.

Moreover, the long-term competitiveness and widespread adoption of immersion cooling technology fundamentally depend on two closely linked frontier areas: advanced coolant material science and comprehensive lifecycle sustainability assessment. The primary task is to develop a new generation of high-performance cooling fluids that exhibit superior thermophysical properties, broad material compatibility, and low global warming potential. At the same time, it is essential to systematically evaluate the chemical compatibility and aging behavior of various coolants with common data center materials—particularly under long-term and high-temperature operating conditions—and to establish an open-access material compatibility database to guide engineering practice. From an environmental perspective, a standardized evaluation framework based on life cycle assessment must be established to quantitatively track the environmental footprint of these systems across all stages—from manufacturing and operation to decommissioning and recycling—thereby providing a scientific basis for the green transformation of this technological pathway.

CRedit authorship contribution statement

Chunjie Yang: Writing – original draft, Formal analysis, Conceptualization. **Xianfeng Zhu:** Writing – review & editing, Supervision, Conceptualization. **Guangyi Zhang:** Formal analysis, Data curation. **Gui Pan:** Project administration, Formal analysis, Data curation. **Xingchuan Wang:** Resources, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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